

EventTrace: Verifying Ordered Semantic Events for Reinforcement Fine-Tuning of Vision-Language Navigation

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Abstract

001 Reinforcement fine-tuning of vision-language navigation
002 (VLN) policies commonly relies on destination suc-
003 cess or similarity to a reference route. These signals
004 leave an instruction’s intermediate requirements unre-
005 solved: reaching the endpoint does not establish that
006 the agent entered the requested room or passed the
007 specified landmark. We study an online reward that
008 represents an instruction as an ordered list of seman-
009 tic events and verifies each event against observations
010 before and after executed actions. A frozen vision-
011 language model supplies a three-way verdict (complete,
012 incomplete, uncertain); motion and stopping checks re-
013 ject visually plausible but unexecuted events. Each
014 event is rewarded at most once, and the process re-
015 ward is bounded below the success bonus. We inte-
016 grate EventTrace into multi-turn ActiveVLN/GRPO
017 and compare it with destination-only training for 64
018 optimization steps across three matched seeds. All six
019 checkpoints and the SFT initialization are evaluated
020 on the same 1,839 R2R val-unseen episodes. Event-
021 Trace changes success rate by +0.65, +0.44, and -2.88
022 percentage points across seeds; its mean paired change
023 is -0.60 points (sample standard deviation 1.98). A
024 single-AI blind review of 49 selected transitions, re-
025 played with training-time image encoding, finds that
026 7 of 15 verifier completion predictions match its la-
027 bels, without independent human adjudication. The
028 method is operational, but these results do not estab-
029 lish a navigation improvement or reliable semantic ver-
030 ification.

031 1. Introduction

032 Vision-language navigation asks an embodied agent to
033 follow a natural-language route through visual obser-
034 vations. The Room-to-Room (R2R) task introduced
035 instruction-guided navigation in real indoor environ-

ments [1]; VLN-CE moves this task into continuous
control [5]. Success rate (SR) and success weighted by
path length (SPL) capture destination attainment, yet
a successful stop can coexist with skipped intermedi-
ate instructions. Instruction fidelity work made this
gap explicit [4].

Recent reinforcement fine-tuning (RFT) approaches
give visual-language policies online or trajectory-level
feedback. ActiveVLN performs multi-turn exploration
with group-relative policy optimization (GRPO) [8,
10]. VLN-R1 uses time-decayed action rewards [7];
Nav-R1 combines format, understanding, and naviga-
tion rewards [6]; SeeNav-Agent proposes step-level ver-
ifiable process rewards in another navigation bench-
mark [9]. Our question is narrower: can a naviga-
tion instruction’s ordered, physical events provide a
bounded reward during closed-loop R2R-CE rollouts,
without matching the policy’s actions to a reference
path?

We introduce EventTrace, a modular reward on top
of an existing multi-turn policy. The reward uses a
short event list, a frozen visual verifier with an absten-
tion state, deterministic checks of motion or stopping,
and a once-only order-preserving tracker. The actor
sees only the original egocentric observations and in-
struction; simulator displacement is available only to
the reward. Figure 1 illustrates the distinction between
seeing a bathroom and actually entering it.

We evaluate the reward with matched three-seed
training and complete R2R val-unseen coverage. The
observed seed variation and an exploratory verifier au-
dit limit the empirical claim: this implementation is
reproducible but does not improve navigation consis-
tently.

2. Related Work

Path and instruction fidelity. SR and SPL reward the
destination and efficiency, whereas nDTW and cover-
age weighted by length score (CLS) compare trajec-

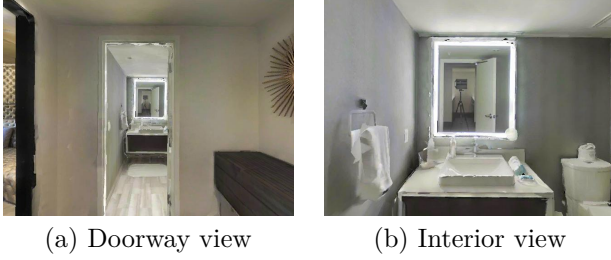


Figure 1. Real simulator frames from a diagnostic replay of R2R training episode 2889. Visibility alone does not complete the event “enter bathroom.” The verifier uses before/after views and executed motion; the ordered tracker later checks “stop near sink.” These frames illustrate the mechanism and are not an independent accuracy test.

ries with a reference route [3, 4]. Reference trajectories can be useful supervision but do not directly answer whether a distinct visual event actually occurred when multiple valid routes exist. Our reward targets event completion during execution; route metrics remain complementary evaluation measures.

RL for VLN. ActiveVLN is our implementation base because its environment can return feedback after each multi-action turn [10]. VLN-R1 and Nav-R1 show other ways to shape VLN rewards [6, 7]. SeeNav-Agent is especially relevant in its emphasis on step-level verification [9]. EventTrace differs in its ordered event state, explicit uncertainty outcome, and physical gates against rewarding a landmark seen without the requested transition. Our pilot still applies GRPO to trajectory-level returns; it does not claim SeeNav-Agent-style step-level credit assignment.

3. Method

3.1. Task and event representation

An instruction x yields a list $E(x) = (e_1, \dots, e_K)$, where e_k contains a verb type, target phrase, and source span. Supported event types include enter, pass, descend, and stop_near. A frozen vision-language model (VLM) proposes at most three events, restricted to source spans in the instruction. The eight-episode wiring pilot uses curated lists; the main 64-step training runs parse instructions automatically at episode start. Parser errors produce an empty list and no semantic reward. We have not independently audited parser accuracy.

In the curated pilot, each of the eight episodes contains a stopping event; there are also three enter, three pass, and three descend events, for 17 in total. These counts describe the pilot distribution, not the preva-

Table 1. Motion gates in the pilot. The VLM must also verify that the image pair supports the specified target.

Event	Count	Required gate
enter	3	Forward action; $\ \Delta p\ \geq 0.10$ m
pass	3	Forward action; $\ \Delta p\ \geq 0.10$ m
descend	3	Vertical change $\Delta y \leq -0.08$ m
stop_near	8	Executed STOP action

lence of event types in R2R. Table 1 shows the deterministic portion of the verifier. Each gate can reject completion before a VLM call, but passing a gate is never sufficient to award a point.

At turn t , the policy receives the instruction, action history, and current egocentric image I_t , and emits up to three low-level actions. The simulator executes these actions and returns I_{t+1} , pose change Δp_t , and task metrics. Only the current pending event e_{k_t} is queried. The frozen verifier produces

$$v_t = V(e_{k_t}, I_t, I_{t+1}, a_t, \Delta p_t) \in \{Y, N, U\}, \quad (1)$$

where U denotes insufficient evidence. A view of a landmark is not by itself a completion verdict. For crossing and passing, a forward displacement check is required; descending requires a negative vertical displacement; stopping near a target requires an executed STOP. The VLM inspects whether the image pair supports the specified target and transition. These are deliberately conservative gates, not a learned proof of physical state.

3.2. Ordered, bounded process reward

Let $k_t \in \{1, \dots, K+1\}$ index the first unfinished event. The event fires only when its verifier says Y. Then

$$c_t = \mathbb{I}[k_t \leq K \wedge v_t = Y], \quad (2)$$

$$k_{t+1} = k_t + c_t, \quad r_t^{\text{sem}} = \frac{\lambda}{K} c_t. \quad (3)$$

A N or U verdict leaves k_t unchanged. The tracker prevents repeated reward for the same landmark, enforces event order, and guarantees $\sum_t r_t^{\text{sem}} \leq \lambda$. Because only the pending event is checked, at most one event fires per turn even if the agent performs several actions. This avoids ambiguous credit when a model emits an action batch but can delay recognition of adjacent events. The tracker retains a confirmed event after later image changes, so a completed event cannot be unconfirmed by viewpoint drift.

The base destination term is the weighted-success reward already used by ActiveVLN, with nDTW weight

zero. We add a fixed success floor b to both arms:

$$R(\tau) = \sum_t r_t^{\text{sem}} + S(\tau) \left[b + \alpha \left(1 - \min \left(\frac{d_T}{3}, 1 \right) \right) \right], \quad (4)$$

where $S(\tau)$ is simulator success after STOP, d_T is terminal distance in meters, $\alpha = 15$, $b = 2$, and $\lambda \in \{0, 1\}$. Hence any success receives at least 2 while a failure receives at most 1 semantic point. This priority holds for this nonnegative reward configuration; it does not imply that the verifier is correct. The bound also makes failure reward interpretable: partial event completion can distinguish two unsuccessful attempts without allowing a failed route to outrank a successful one solely through event bonuses. The STOP event is distinct from simulator success: it demands visual proximity to its instruction target, while $S(\tau)$ uses the simulator’s goal region. Disagreement between these signals is precisely a case the proposed reward aims to expose.

For each training episode, we sample two trajectories and normalize their returns within the group for GRPO [8]. Event rewards are emitted after each turn, but the current trainer sums them before constructing a trajectory advantage. Consequently, the implementation changes which trajectories are preferred, not which individual turn receives the policy gradient. Turn-level event timing is available in the log for future credit-assignment work.

3.3. Verifier failure policy

The verifier is a frozen Qwen3-VL-8B-Instruct model. A service timeout, malformed response, or uncertain decision yields zero event reward; it does not end the episode. The reward requires no reference action prefix. Simulator pose is used only for reward verification, never added to the actor observation. In the pilot, the fixed event manifest is matched to both episode ID and exact instruction text to avoid assigning another episode’s event list; the main runs use online parsing. The verifier sees two RGB frames resized to fit within 448×448 pixels, the event type and target, its source phrase, the executed action text, translation magnitude, and vertical displacement. Its output is a deterministic one-letter Y/N/U response; generated explanations are not treated as confidence scores. This makes the reward externally auditable through a saved per-turn verdict, although a forced-choice label is still fallible. The HTTP verifier is separated from the simulator so it can be frozen and logged independently of policy updates.

4. Experimental Setup

Environment and policy. We use R2R train in VLN-CE, rendered from Matterport3D scans [2], the ActiveVLN implementation, and an SFT-initialized Qwen2.5-VL-3B policy. Matterport3D data and derived imagery are subject to its [academic terms of use](#). Each turn contains at most three grouped action commands. The configured limits are 12 turns and 36 commands; the strict budget check can permit one extra command, and a command may execute several simulator steps. Actor sampling uses temperature 1.2 and two independently sampled rollouts per item. GRPO trains on two A800 GPUs at learning rate 10^{-6} with batches of four episodes. The destination-only control sets $\lambda = 0$; the event arm sets $\lambda = 1$. Both arms have the same SFT initialization, destination reward, data order within seed, and optimizer budget.

Pilot and corrected main training. We first replay eight training episodes with expert actions solely to inspect expected events, then run a two-step reward-wiring pilot with curated events. These replays are neither held-out labels nor training results for the main comparison. For the main study, each arm runs 64 optimization steps at seeds 11, 22, and 33: 256 train-episode exposures and 512 sampled rollouts per arm and seed. The event arm uses automatically parsed events. A preflight check detected that a fixed per-request sampling seed duplicated every two-rollout GRPO group and yielded zero actor gradients; we excluded that attempt, removed the per-request seed while retaining the engine seed, and reran all six models. Every corrected group has distinct sampled actions. The archive records the excluded attempt and corrected patch.

Full val-unseen evaluation. SR is the fraction of trajectories satisfying the simulator goal condition; SPL also accounts for path efficiency. Mean terminal goal distance is in meters. Training rollout success and return are diagnostics, not held-out outcomes; returns differ in definition between arms. The SFT model and six corrected step-64 checkpoints run on the identical manifest of all 1,839 R2R val-unseen episodes from 11 scenes. We use one generation per episode, temperature 0.2, top- p 0.8, a 12-turn limit, and at most 76,800 image pixels. Four Habitat shards per checkpoint divide the manifest without overlap. The evaluator uses the same prompt and environment-specific OpenAI-compatible request and image-resize shim for all seven models. The completion gate checks the exact episode-ID set and zero inference errors. Episode IDs are paired

Table 2. Observed pilot results on eight training episodes. Each arm ran two optimizer steps and produced 16 sampled rollouts. The event arm’s larger mean return is expected from the added reward and does not indicate better navigation.

Metric	Destination only	EventTrace
Successful rollouts	13/16	13/16
Mean trajectory return	9.325	10.455
Mean event progress	–	0.656
Confirmed event hits	–	23
Uncertain verifier turns	–	33
Verifier service errors	–	0

within each seed, but stochastic decoding uses only one generation per checkpoint and is an additional source of variation.

Blind verifier review. Separately, we collect shortest-path follower trajectories from 16 different val-unseen episodes, of which 12 yield at least one automatically parsed event. Before querying the verifier, one AI-assisted reviewer inspects RGB image pairs, executed actions, and motion and fixes 49 three-way completion labels across those 12 episodes. For the reported replay, both frames use the training client’s 448-pixel JPEG encoding. The sample includes apparent transitions, controls, and final stops. Labels are independent of verifier predictions, but there is no second reviewer or human ground-truth adjudication, and the selected class proportions are not representative of deployment traffic. We therefore treat this as a failure analysis, not a population accuracy estimate.

5. Results and Analysis

Reward reaches optimization. In the first optimizer step, both arms sample the same eight action sequences and attain seven successes. For every matched trajectory, the difference in total return equals its semantic event score, including a failed episode 7110 trajectory that receives 0.5 for completing “descend stairs” but never reaches the destination. The modified return is consumed by the training reward manager, and both arms save step-2 checkpoints after nonzero policy updates. This verifies the wiring of Eq. 4; it is not evidence that event reward improves the policy.

What the verifier recognizes. In diagnostic expert replays, four of eight episodes receive full event credit, two receive partial credit, and two receive none; the tracker confirms 11 of 17 curated events. For episode 2889, seeing the sink from outside the bathroom produces uncertain verdicts before entrance, while later

Table 3. Complete R2R val-unseen evaluation. Every row uses the same 1,839 episodes from 11 scenes, one generation per episode, and zero inference errors. Goal distance is in meters. C: destination-only control; E: EventTrace.

Checkpoint	Successes	SR (%)	SPL (%)	Dist.
SFT	544	29.58	28.69	7.03
Seed 11 C	479	26.05	25.32	7.75
Seed 11 E	491	26.70	25.91	7.63
Seed 22 C	566	30.78	30.14	6.91
Seed 22 E	574	31.21	30.63	6.98
Seed 33 C	502	27.30	26.65	7.14
Seed 33 E	449	24.42	23.16	7.76

motion and view changes confirm enter. A rotation followed by premature STOP yields zero semantic credit in a negative replay. There are also false negatives: one shallow stair descent is rejected by the vertical-motion threshold, and one landmark is never confirmed. Because the events were curated using these episodes and the replays lack independent annotations, 11/17 must not be reported as verifier recall or accuracy.

Main training diagnostics. All six corrected runs finish 64 steps, with 256 diverse GRPO groups per arm and seed. Training-rollout success is 111/512 versus 119/512 (control versus event) for seed 11, 108/512 versus 119/512 for seed 22, and 88/512 versus 82/512 for seed 33. These on-policy samples change as training proceeds, so they are not comparable to the held-out evaluation. The event verifier records 272, 275, and 243 confirmed event hits over the three event-arm runs. These counts are not event accuracy: events are automatically parsed and no independent labels cover these rollouts.

Complete val-unseen results. Table 3 gives all seven checkpoints on the same 1,839 episodes. Within matched seeds, EventTrace changes SR by +0.65, +0.44, and −2.88 percentage points, and SPL by +0.58, +0.49, and −3.49 points. The mean paired seed change is -0.60 ± 1.98 points for SR and -0.80 ± 2.32 points for SPL (mean $\pm$ sample standard deviation over three seeds). At the episode level, event-only versus control-only successes are 157 versus 145, 197 versus 189, and 155 versus 208. This variation does not support a consistent improvement. All seven evaluations have exact 1,839-episode coverage and zero logged inference errors. The earlier 16-episode, two-step probe yielded 6/16 successes for every arm; it was exploratory and is superseded by the complete comparison.

Table 4. Training-encoded replay of 49 selected val-unseen event transitions from 12 episodes. Rows are blind review labels and columns are frozen verifier predictions. U means evidence insufficient; the single-AI review has no independent human adjudication.

	Pred. Y	Pred. N	Pred. U
Review Y	7	0	1
Review N	4	13	12
Review U	4	1	7

Verifier failure analysis. Table 4 compares the frozen verifier with labels fixed before its predictions were obtained. Seven of eight reviewed completions receive Y, but only 7 of 15 Y predictions match a reviewed completion; four are on clear non-completions and four on ambiguous turns. One false positive occurs after the agent has already entered the requested room, exposing a failure to distinguish current state from an event newly completed this turn. Motion gates alone do not eliminate this error. These selected examples cannot establish deployment precision, but they make verifier reliability a concrete limitation of the present reward.

6. Discussion and Limitations

The main risk is reward misspecification: the VLM may confuse doorways, and displacement gates may miss small valid movements. One-at-a-time verification may miss events completed together. Before/after frames limit timing to a turn; trajectory-level GRPO offers limited credit assignment to the decisive action.

The complete val-unseen comparison shows a negative mean effect and large seed variation, but 64 steps expose only 256 training-episode instances per run. For EventTrace, a larger training budget and repeated decoding have not been evaluated. The selected 49-example review has one AI labeler and no human adjudication; instruction parsing is also unaudited. We cannot isolate noisy verification, parser errors, reward scale, budget, or optimization as the cause. Human-labeled verifier and parser sets, followed by ablations of ordering, abstention, gates, reward budget, and turn-level advantage, are needed. This audit cannot establish event precision or recall.

Exploratory policy-prefix follow-up. Separately from EventTrace, we tested whether changing the training-state distribution helps destination-only GRPO. For each of three seeds, a branching curriculum replayed a training-only policy prefix before sampling two continuations; its matched control started at the task origin. Both arms used the same 512 train episodes, SFT

Table 5. Exploratory 128-step policy-prefix follow-up on all 1,839 val-unseen episodes. Differences are branching minus a same-data, from-scratch destination-only control, in percentage points. Mean $\pm$ sample standard deviation is over three training seeds; each model has one stochastic decode per episode.

Seed	Δ SR	Δ SPL
11	-7.18	-6.86
22	+0.49	+0.80
33	+3.26	+4.41
Mean $\pm$ SD	-1.14 ± 5.41	-0.55 ± 5.75

initialization, destination reward, and 128 optimizer steps. All six checkpoints covered the complete 1,839-episode val-unseen split with zero inference errors. Table 5 shows sharply different seed effects and a negative three-seed mean. Exploratory scene-and-seed 95% intervals for the mean differences are $[-6.51, 4.02]$ SR and $[-6.16, 4.85]$ SPL points. On the 1,583 episodes outside the fixed 256-episode development screen, the mean changes remain -1.22 SR and -0.55 SPL points. Separate stochastic decodes of the overlapping 256 episodes changed success outcomes for 20–41 episodes per checkpoint, so the small screen is an unstable model-selection signal. This follow-up changes start state and remaining budget together; it does not identify why EventTrace failed or establish a benefit from branching. The checked paired episode records and training audits accompany the manuscript.

Group-four optimizer follow-up. A separate destination-only comparison tests four sampled trajectories per GRPO group against the same-data, two-sample control. Both arms use the same SFT initialization, 512 unique training episodes, and 128 optimizer steps at each of three matched seeds; the four-sample arm uses twice as many simulator rollouts. All six checkpoints cover the identical 1,839-episode val-unseen manifest with zero inference errors. Table 6 shows positive paired differences in each seed: the mean is $+2.10 \pm 1.93$ SR and $+2.51 \pm 2.27$ SPL percentage points (mean $\pm$ sample standard deviation over seeds). On the 1,583 episodes outside the fixed 256-episode screen, the means are $+1.94$ SR and $+2.40$ SPL points. The largest seed-33 gain is $+4.30$ SR points, while exploratory 11-scene bootstrap intervals for seeds 11 and 22 cross zero. One stochastic decode per checkpoint and only three training seeds limit the inference. This comparison does not identify a better semantic reward; subsequent reward comparisons must hold group size at four in both arms. The complete paired episode records, training audits, and

Table 6. Exploratory 128-step, destination-only group-size follow-up on all 1,839 val-unseen episodes. Differences are four samples minus a matched two-sample control, in percentage points; the four-sample arm uses twice the simulator rollouts.

Seed	Δ SR	Δ SPL
11	+0.65	+0.96
22	+1.36	+1.44
33	+4.30	+5.12
Mean $\pm$ SD	$+2.10 \pm 1.93$	$+2.51 \pm 2.27$

independent recomputation script accompany the paper.

Instruction-grounded reward representation. We also probe a potential reward that combines a causal progress encoder from frozen navigation-policy states with a SigLIP image-instruction encoder adapted on training-scene goal panoramas and group-four successful/failed routes. SigLIP text is padded to a fixed 64 tokens, and offline scoring uses the same one-image request shape as the online service. Each component is normalized by its fit-scene mean absolute margin; a fixed 1:1 mixture is applied without development-set tuning. On 52 development pairs, successful-route ranking rises from 37/52 for the temporal term and 32/52 for the visual term to 43/52 for the mixture; correct-versus-wrong-instruction preference is 39/52. On 38 new episode-disjoint, train-scene pairs, the mixture scores 30/38 route ranks and 34/38 instruction comparisons. A previously inspected 48-pair scene audit scores 41/48 and 37/48. These selected offline pairs share train scenes or prior audit use, so they provide a screen for an online group-four reward comparison, not evidence of val-unseen navigation gain.

Online fused-reward screen. We added this frozen potential as a bounded terminal bonus to the destination reward and trained a matched group-four policy for 64 steps at seed 11. Its training audit verifies 1,024 reward requests for 1,024 rollouts across 256 matched train episodes. On the fixed 256-episode val-unseen screen, the fused-reward checkpoint achieves 68/256 successes versus 80/256 for the same-data, same-group-size destination-only control: paired SR changes by -4.69 percentage points and SPL by -4.34 points. Both arms cover exactly 256 episodes in 11 scenes with zero inference errors. This single-seed screen does not support scaling this particular reward, despite its offline ranking results. A failure-aware encoder trained on unsuccessful pairs improves a separate 106-pair train-scene ranking audit from 69/106 to 79/106, but

direct fusion with the image-instruction score weakens reused instruction-matching probes. Those representation findings do not establish a navigation improvement.

Failure-only reward screen. We next applied the new temporal potential only to unsuccessful rollouts, leaving successful rollouts on the original destination reward. A matched seed-11 group-four policy completed 64 steps on the same training episodes; its audit found 854 reward requests for 854 unsuccessful rollouts and no extra bonus for 170 successful rollouts. On the same fixed 256 val-unseen episodes, it achieves 63/256 successes versus the destination-only control's 80/256: paired SR changes by -6.64 points and SPL by -6.05 points. Both models cover 11 scenes without inference errors. The exploratory scene-cluster 95% intervals are $[-11.07, -2.29]$ SR and $[-10.66, -1.47]$ SPL points. This screen fails the predeclared positive SR-and-SPL scale gate; we therefore do not run a three-seed scale-up for this reward. Training-only failed-pair ranking (622/970 in all-failure groups) does not transfer into a navigation benefit. A post-hoc behavior check finds maximum-turn termination on 187 candidate episodes versus 107 control episodes, while mean traveled path length falls from 6.74 to 5.71 m; these differences motivate a stopping-aware reward test but do not identify the cause of the regression. An offline equal mixture with the adapted visual score also fails its train-scene development gate (68/125 failed-pair ranks versus 90/125 for the temporal term), so its audit remains unopened.

Stop-aware reward screen. To test whether terminal-mode conditioning mitigates this failure, we gave the frozen temporal bonus only to unsuccessful voluntary STOP trajectories. Successes retained the destination reward, while turn-limit and format failures received no bonus. The matched seed-11 group-four policy completed 64 steps on the same training episodes. Its audit found exactly 328 scorer requests for 328 unsuccessful voluntary stops across 1,024 rollouts, with nonzero actor gradients at every step. On a new, disjoint 256-episode val-unseen screen covering 10 scenes, the candidate succeeds on 45/256 episodes versus 93/256 for its same-seed destination-only control. Paired SR and SPL changes are -18.75 and -17.43 percentage points; both arms have zero inference errors. The predeclared positive SR-and-SPL gate failed, so this reward received no three-seed expansion. Post-hoc analysis finds mean path length 4.44 versus 6.93 m and only three candidate episodes with a non-null termination reason versus 116 controls, consistent with frequent early

stops. On the candidate’s train rollouts, its frozen raw score has AUC 0.486 for distinguishing 18 unsuccessful STOP endpoints within 3.5 m of the goal from 310 farther endpoints; only 11 of 183 positive-bonus stops are within 3.5 m. This train-scene diagnostic does not establish causality, but it shows that failure-pair ranking is insufficient calibration for rewarding STOP.

7. Conclusion

We formulated a path-free semantic process reward for closed-loop VLN that verifies ordered instruction events against executed motion and before/after views. A wiring pilot confirms that the bounded reward enters online GRPO. In three matched 64-step training runs and complete 1,839-episode val-unseen evaluation, EventTrace has a mean paired SR change of -0.60 percentage points relative to destination-only training, with inconsistent seed effects. A single-AI blind review exposes verifier disagreement but provides no independent ground truth. The current method is reproducible; reliable semantic verification and a navigation benefit remain unproven.

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